In [ ]:
```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk import pos_tag
# Download required NLTK resources (run once)
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('averaged_perceptron_tagger_eng')
# Download once
nltk.download('averaged_perceptron_tagger_eng')
```

NLP part 1 Tokenization

In NLP, tokenization means breaking text into smaller pieces called tokens. The main types are:

1. Word tokenization — splits text into words.

| Type | Example | |---|---| | Word | `"I love Python"` → `["I", "love", "Python"]` | | Sentence | `"Hi. How are you?"` → `["Hi.", "How are you?"]` | | Character | `"cat"` → `["c", "a", "t"]` | | Subword | `"playing"` → `["play", "ing"]` |

In [41]:
```
text = "Python is a simple and powerful programming language for data processing."
# 1. Tokenization first way
tokens = word_tokenize(text)
print("Tokens:")
print(tokens)
```

```
Tokens:
['Python', 'is', 'a', 'simple', 'and', 'powerful', 'programming', 'language', 'for', 'data', 'processing', '.']
```

In [44]:
```
# 2 . tokenization second way
#for word in text.split(" "):
    #print(word)

text2 = "Python, is, a, simple, and, powerful, programming, language, for, data, processing."
```

```
for word in text2.split(","):
    print(word)
```

```
Python
 is
 a
 simple
 and
 powerful
 programming
 language
 for
 data
 processing.
```

Tokenization 2 Sentence tokenizer

In [45]:
```
from nltk.tokenize import sent_tokenize

text3 = "I love Python. NLP is interesting."
sentences = sent_tokenize(text3)
print(sentences)
```

```
['I love Python.', 'NLP is interesting.']
```

Tokenization 3 Character tokenizer

In [46]:
```
text4 = "Python"

tokens4 = list(text4)
print(tokens4)
```

```
['P', 'y', 't', 'h', 'o', 'n']
```

In [50]:
```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")
```

```
text5 = "unhappy"

tokens5 = tokenizer.tokenize(text5)

print(tokens5)
```

```
['unhappy']
```

Type	Exercise
Word	Split `"Python is easy to learn"` into words
Sentence	Split `"Python is easy. I like Python. NLP is interesting."` into sentences
Character	Split `"NLP"` into individual characters
Subword	Break `"unhappiness"` into smaller meaningful parts

NLP Part 2 Stopwords

Stopwords are very common words in a language that often carry little useful information for many NLP tasks. Examples: the, is, a, an, and, in, of, to, for

In [59]:
```
stop_words = set(stopwords.words('english'))
stop_words_finnish = set(stopwords.words('swedish'))


text = "Python is a simple and powerful programming language for data processing."

#stop_words_finnish
```

In [57]:
```
# 1. Tokenization first way
tokens = word_tokenize(text)
filtered_tokens = [
    word for word in tokens
    if word.lower() not in stop_words and word.isalpha()
]
```

```
print("\nAfter Stopword Removal:")
print(filtered_tokens)
```

```
After Stopword Removal:
['Python', 'simple', 'powerful', 'programming', 'language', 'data', 'processing']
```

NLP part 3 Parts of Speech Taggings

In [60]:
```
# 3. POS Tagging
pos_tags = pos_tag(filtered_tokens)

print("\nPOS Tags:")
print(pos_tags)
```

```
POS Tags:
[('Python', 'NNP'), ('simple', 'JJ'), ('powerful', 'JJ'), ('programming', 'JJ'), ('language', 'NN'), ('data', 'NNS'), ('process
ing', 'NN')]
```

In [61]:
```
# Example text
text = """
The intelligent student writes beautiful programs and analyzes large datasets.
She develops powerful applications and creates useful solutions.
The experienced teacher explains difficult concepts and gives practical examples.
"""

# 1. Tokenization
tokens = word_tokenize(text)

# 2. Remove stopwords and punctuation
stop_words = set(stopwords.words('english'))

filtered_tokens = [
    word for word in tokens
    if word.lower() not in stop_words and word.isalpha()
]

print("Filtered Tokens:")
print(filtered_tokens)
```

```
# 3. POS Tagging
pos_tags = pos_tag(filtered_tokens)

print("\nPOS Tags:")
print(pos_tags)

# 4. Separate Nouns, Verbs, and Adjectives

nouns = []
verbs = []
adjectives = []

for word, tag in pos_tags:

    # Nouns: NN, NNS, NNP, NNPS
    if tag.startswith('NN'):
        nouns.append(word)

    # Verbs: VB, VBD, VBG, VBN, VBP, VBZ
    elif tag.startswith('VB'):
        verbs.append(word)

    # Adjectives: JJ, JJR, JJS
    elif tag.startswith('JJ'):
        adjectives.append(word)

print("\nNouns:")
print(nouns)

print("\nVerbs:")
print(verbs)

print("\nAdjectives:")
print(adjectives)
```

```
Filtered Tokens:
['intelligent', 'student', 'writes', 'beautiful', 'programs', 'analyzes', 'large', 'datasets', 'develops', 'powerful', 'applica
tions', 'creates', 'useful', 'solutions', 'experienced', 'teacher', 'explains', 'difficult', 'concepts', 'gives', 'practical',
'examples']

POS Tags:
[('intelligent', 'JJ'), ('student', 'NN'), ('writes', 'VBZ'), ('beautiful', 'JJ'), ('programs', 'NNS'), ('analyzes', 'VBP'),
('large', 'JJ'), ('datasets', 'NNS'), ('develops', 'VBZ'), ('powerful', 'JJ'), ('applications', 'NNS'), ('creates', 'VBZ'), ('u
seful', 'JJ'), ('solutions', 'NNS'), ('experienced', 'VBD'), ('teacher', 'NN'), ('explains', 'NNS'), ('difficult', 'JJ'), ('con
cepts', 'NNS'), ('gives', 'VBZ'), ('practical', 'JJ'), ('examples', 'NNS')]

Nouns:
['student', 'programs', 'datasets', 'applications', 'solutions', 'teacher', 'explains', 'concepts', 'examples']

Verbs:
['writes', 'analyzes', 'develops', 'creates', 'experienced', 'gives']

Adjectives:
['intelligent', 'beautiful', 'large', 'powerful', 'useful', 'difficult', 'practical']
```